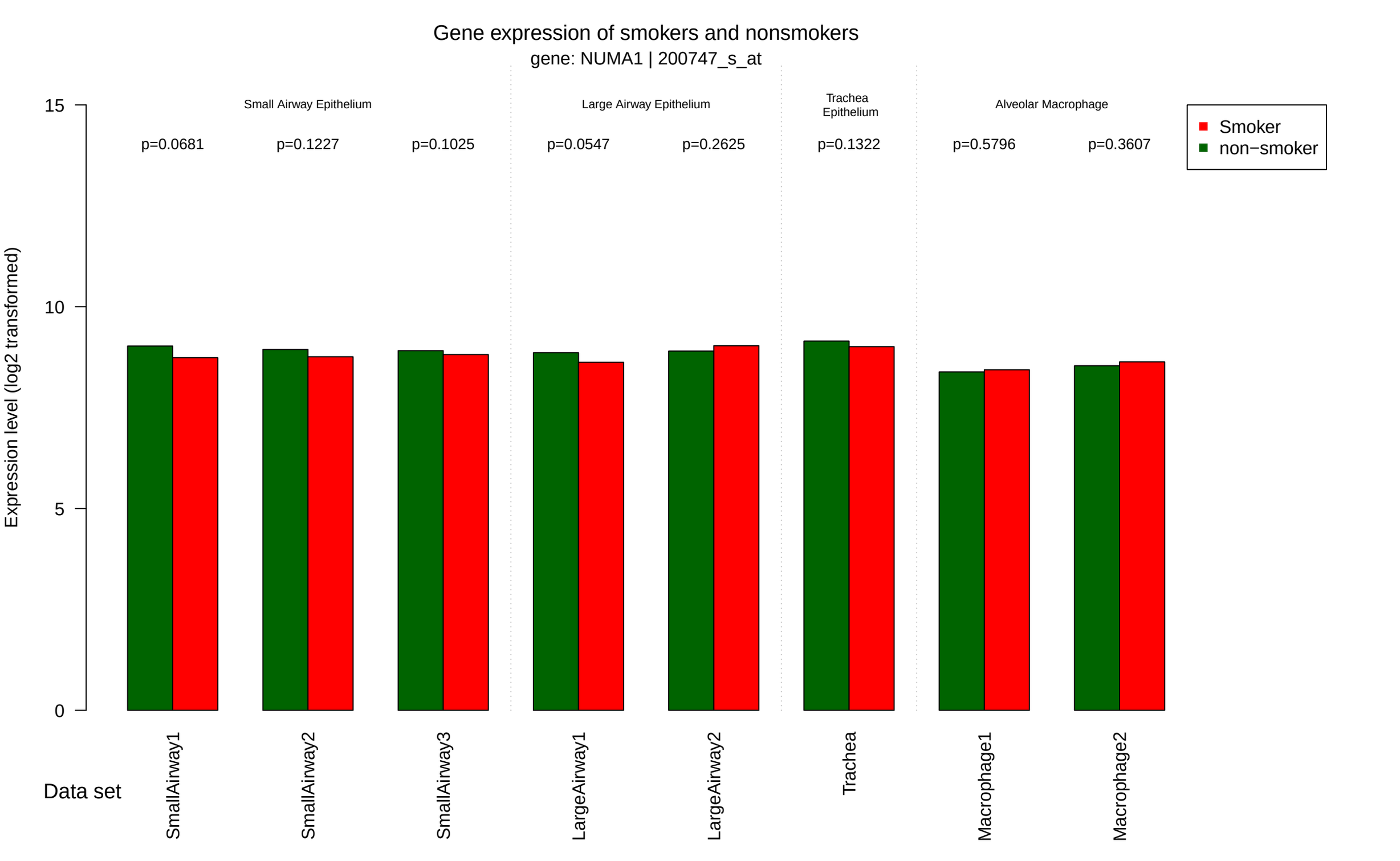
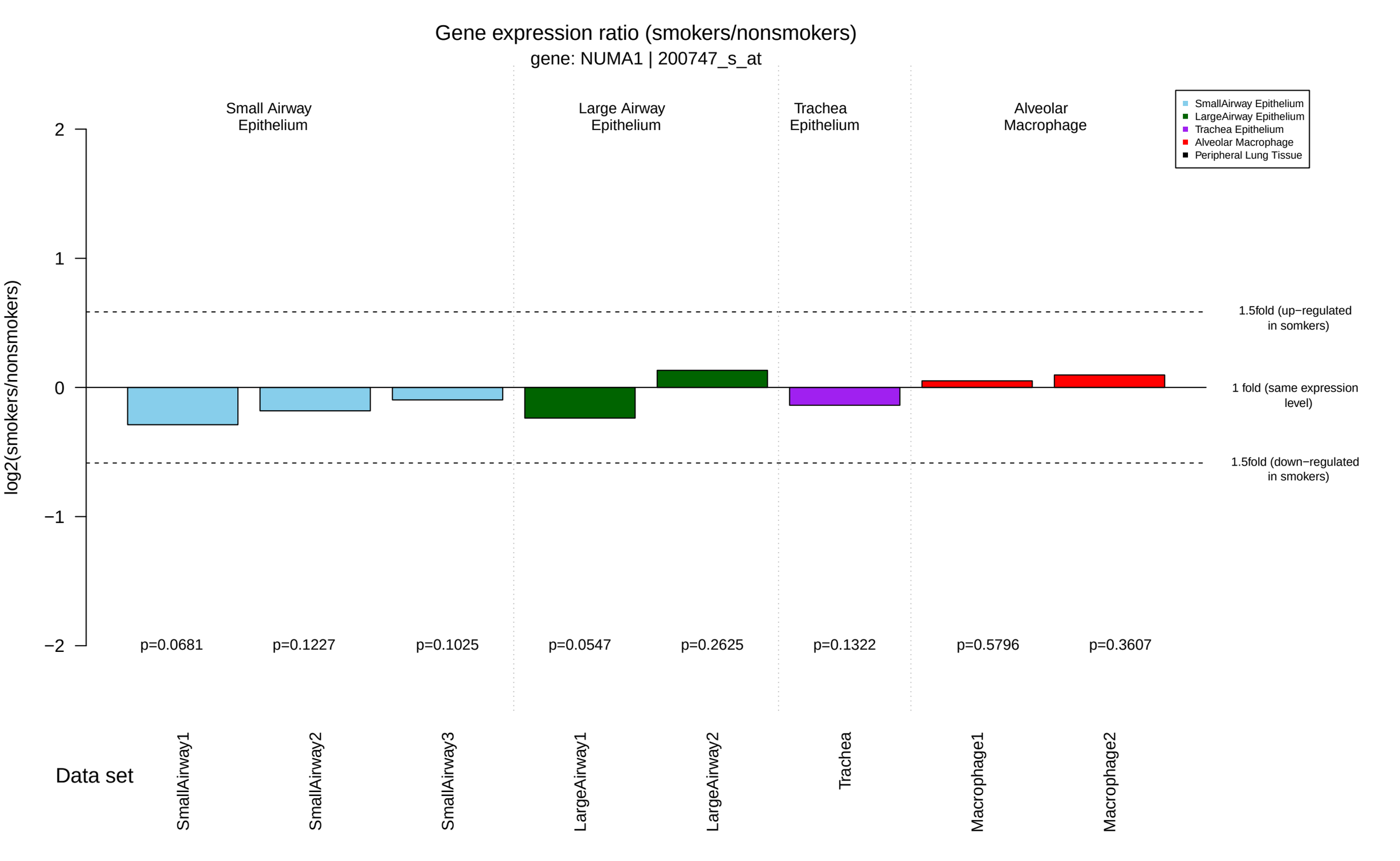
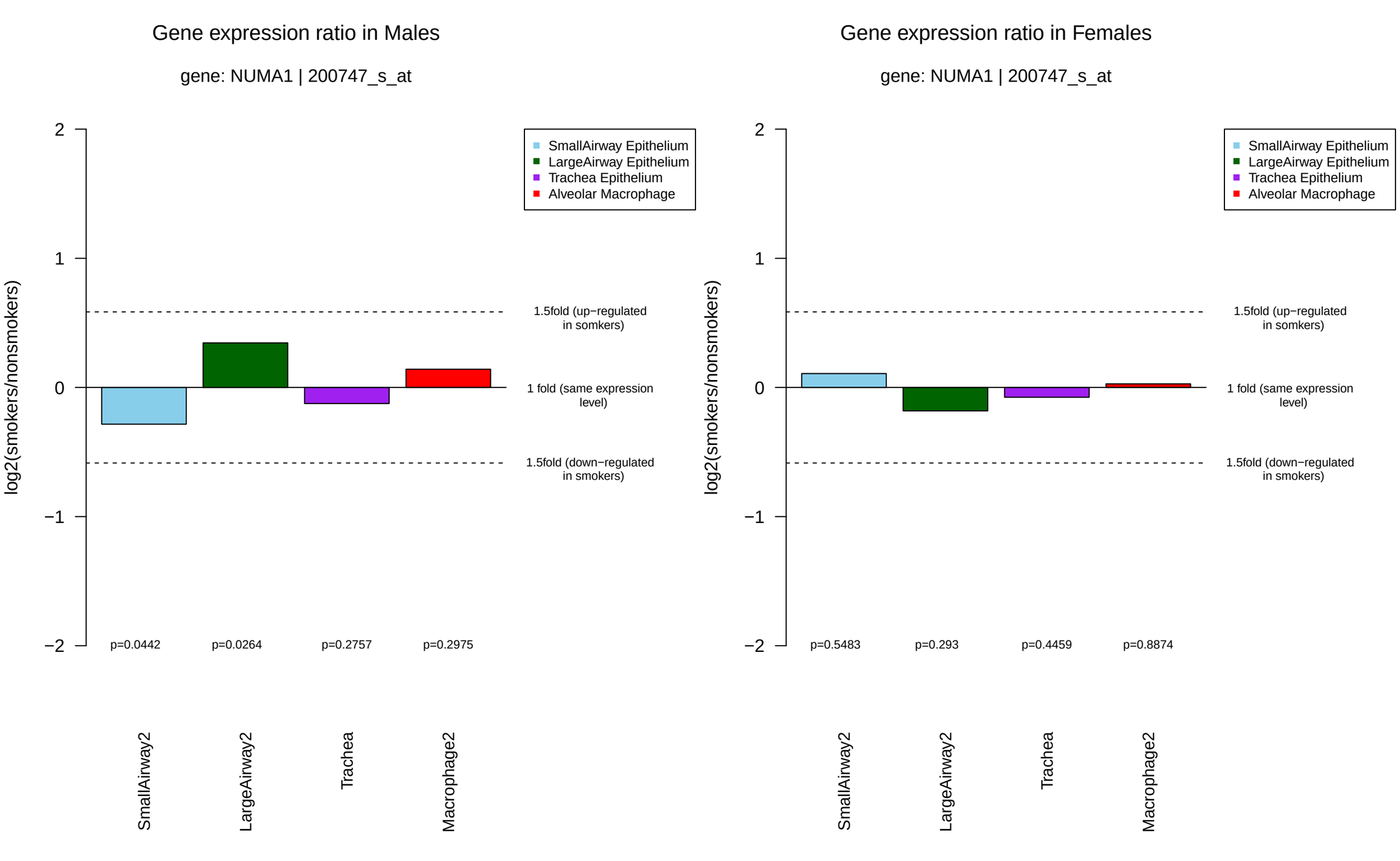
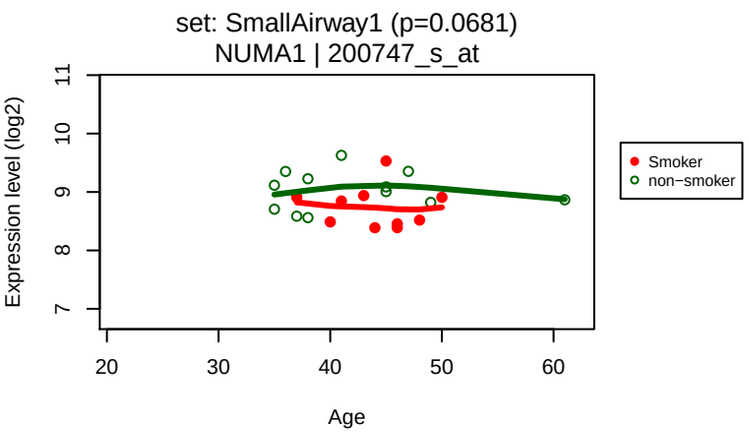
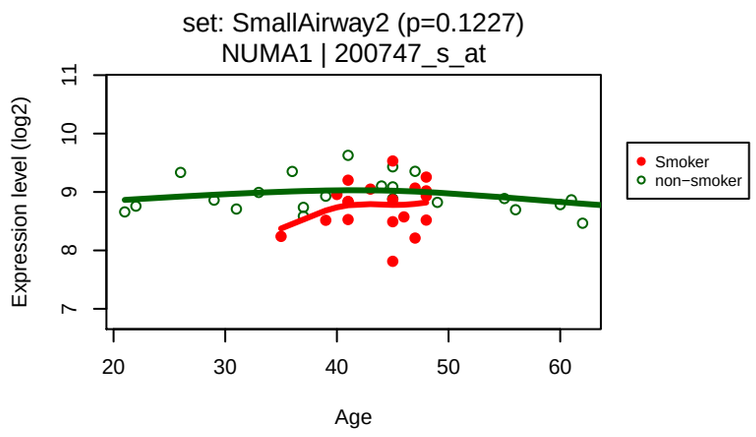
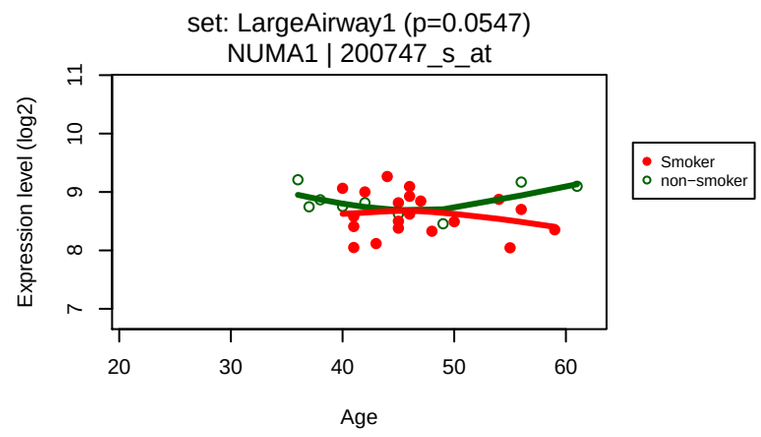

Gene expression of smokers and nonsmokers
gene: NUMA1 | 200747_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0681
p=0.1227
p=0.1025
p=0.0547
p=0.2625
p=0.1322
p=0.5796
p=0.3607
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

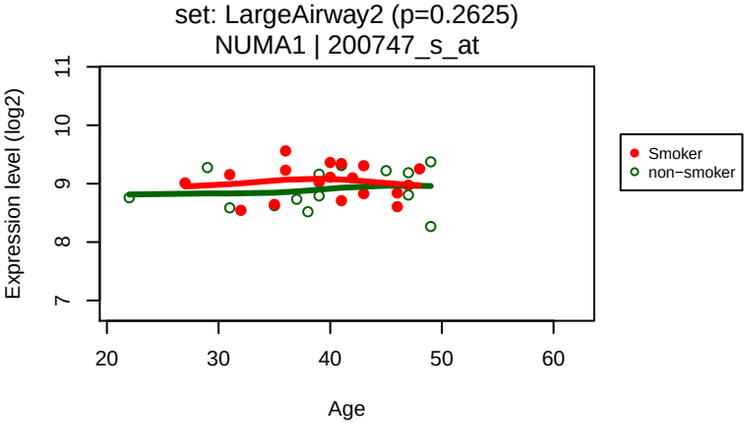

Gene expression ratio (smokers/nonsmokers)
gene: NUMA1 | 200747_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0681
p=0.1227
p=0.1025
p=0.0547
p=0.2625
p=0.1322
p=0.5796
p=0.3607
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

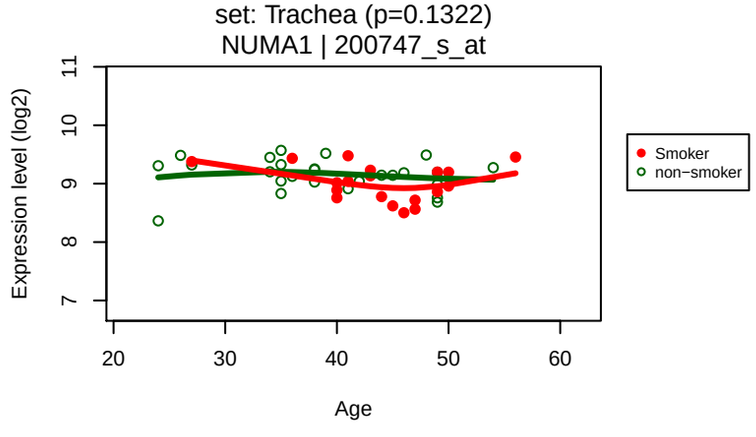

Gene expression ratio in Males
gene: NUMA1 | 200747_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0442
p=0.0264
p=0.2757
p=0.2975
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NUMA1 | 200747_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5483
p=0.293
p=0.4459
p=0.8874
SmallAirway2
LargeAirway2
Trachea
Macrophage2

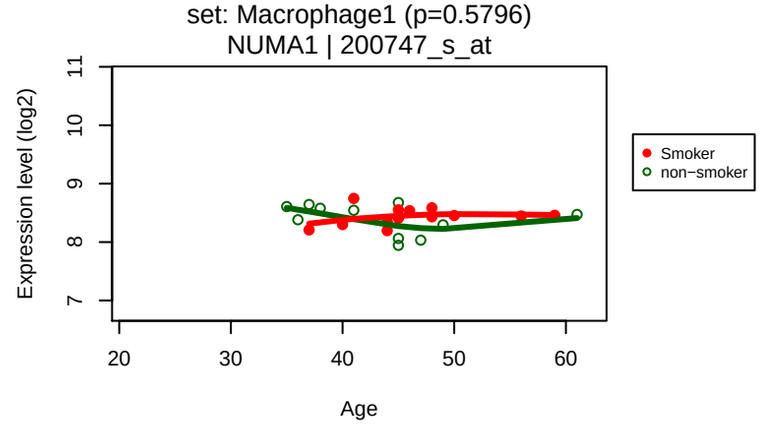

set: SmallAirway1 (p=0.0681)
NUMA1 | 200747_s_at
Expression level (log2)
Age
Smoker
non−smoker

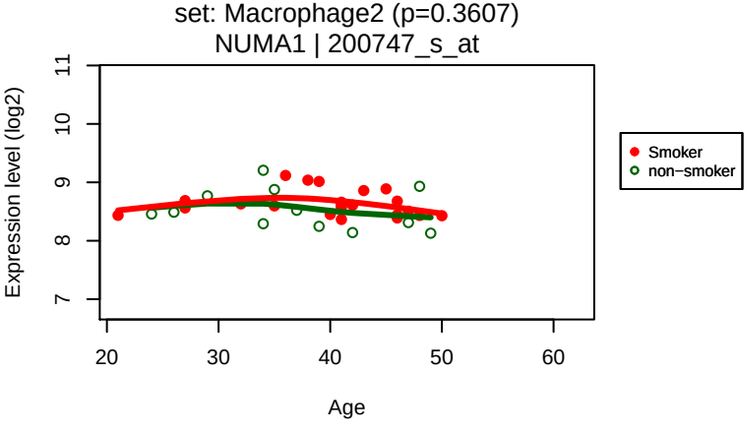

set: SmallAirway2 (p=0.1227)
NUMA1 | 200747_s_at
Expression level (log2)
Age
Smoker
non−smoker

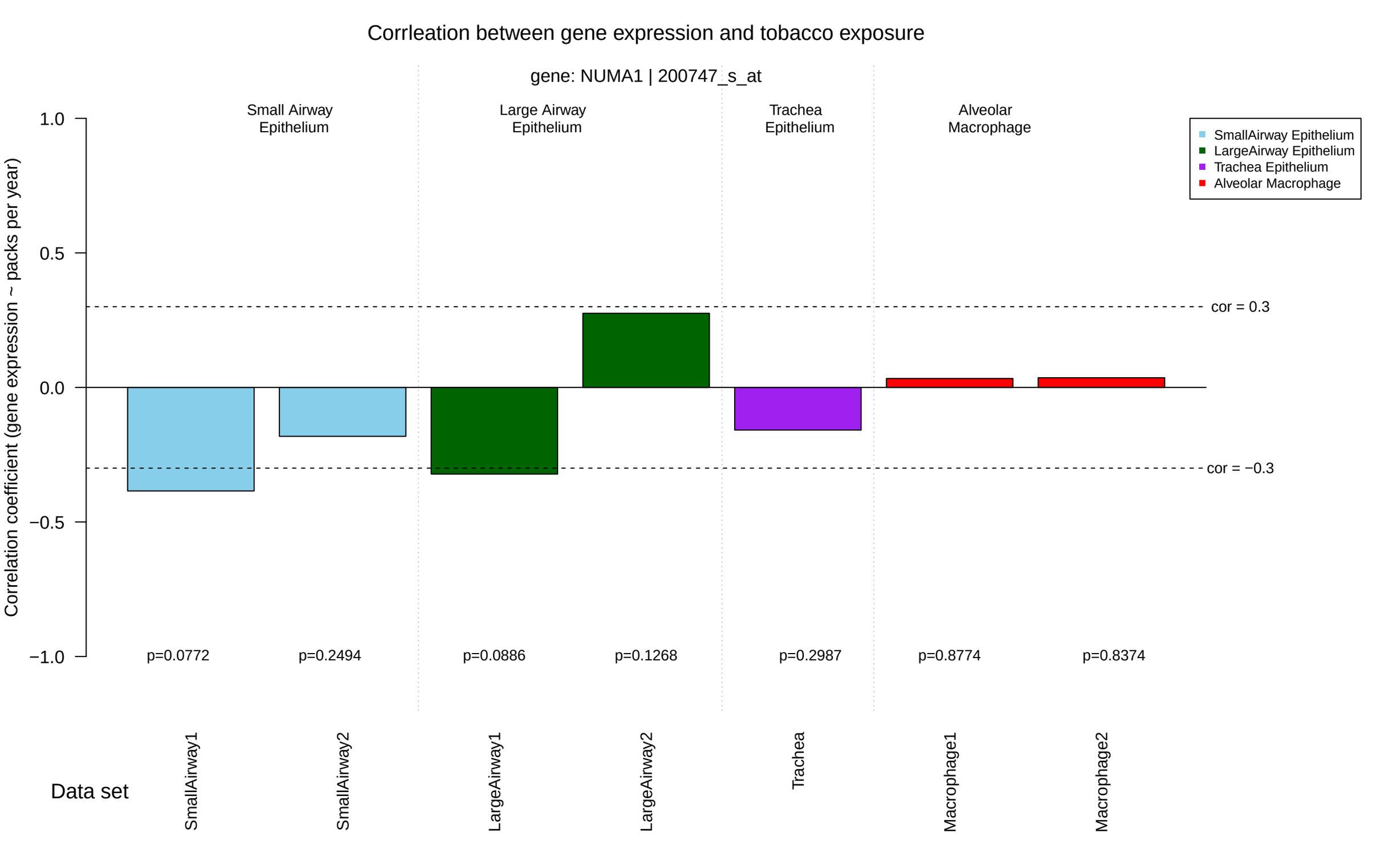

set: LargeAirway1 (p=0.0547)
NUMA1 | 200747_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2625)
NUMA1 | 200747_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1322)
NUMA1 | 200747_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5796)
NUMA1 | 200747_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3607)
NUMA1 | 200747_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NUMA1 | 200747_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0772
p=0.2494
p=0.0886
p=0.1268
p=0.2987
p=0.8774
p=0.8374
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2